

DEEP LEARNING CAPSTONE REPORT

Benchmarking Feedforward and Convolutional Neural Networks
for Handwritten Digit Classification on the MNIST Dataset

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Assessment Type: Final Project Submission (Option 1)

1. Executive Summary & Project Objective

The main objective of this analysis is to evaluate and compare multiple deep learning models to perform high-accuracy image classification. Specifically, this project benchmarks three distinct supervised deep learning architectures on the MNIST handwritten digit dataset: a Multi-Layer Perceptron (MLP), a simple Convolutional Neural Network (CNN), and a regularized CNN (featuring Batch Normalization and Dropout). By training and comparing these models, we seek to identify the most accurate and explainable model that fits deployment needs.

Business and Stakeholder Benefits: This analysis demonstrates key technical methodologies in designing automated optical character recognition (OCR) engines. In commercial applications, accurate digit classification is critical for bank check processing, postal mail sorting, and invoice digitization. Implementing an optimal architecture ensures minimal manual validation cost, faster processing throughput, and reduced error rates in data ingestion pipelines.

2. Dataset Description

The analysis was performed on the classic **MNIST dataset**, a subset of a larger database available from NIST. The dataset contains a total of 70,000 grayscale images of handwritten digits (0 through 9).

Attribute	Value / Description
Total Dataset Size	70,000 samples (60,000 training, 10,000 testing)
Image Dimensions	28 x 28 pixels (Grayscale, 1 channel)
Target Classes	10 classes (digits 0 to 9)
Feature Scaling	Normalized to range [0, 1] by dividing by 255.0
Validation Split	20% of training data used for validation

3. Preprocessing & Model Architectures

Before modeling, the pixel values of the images (originally 0-255) were normalized to the range [0.0, 1.0] to facilitate stable gradient descent. The dataset was split into 48,000 training images, 12,000 validation images, and 10,000 test images.

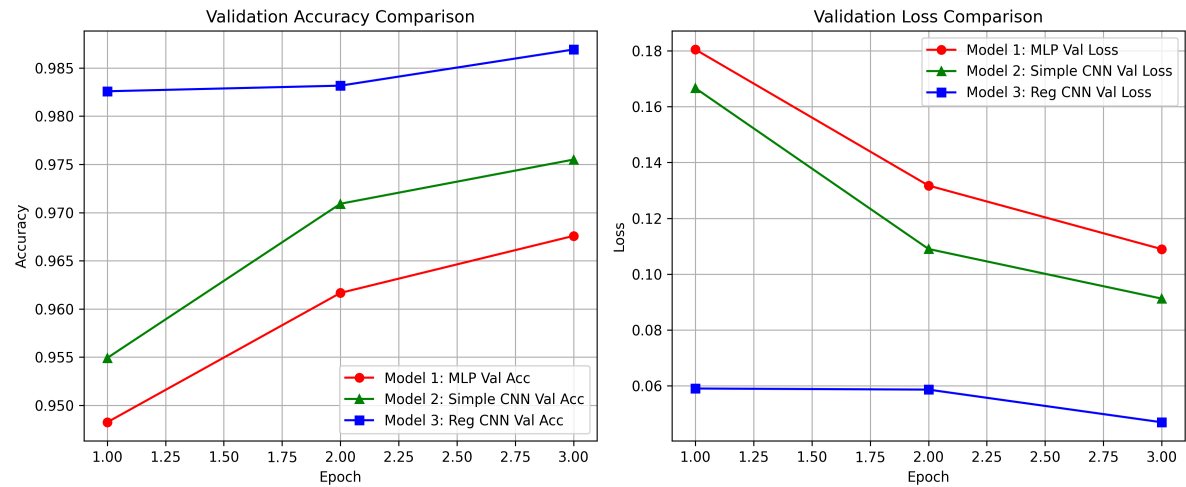
Three model variations were trained to benchmark performance:

- **Model 1 (Simple MLP):** A basic feedforward neural network. It flattens the 28x28 image into a 784-dimensional vector, passes it through a Dense hidden layer (128 units, ReLu activation), and connects to a 10-unit Softmax output layer.

- **Model 2 (Simple CNN):** A convolutional network that leverages spatial structure. It features a Conv2D layer (16 filters, 3x3 kernel size, ReLu activation) followed by a MaxPooling2D layer (2x2) and a 10-unit Softmax output layer.
- **Model 3 (Regularized CNN):** An advanced CNN containing two Conv2D layers (32 and 64 filters respectively with 3x3 kernels), Batch Normalization layers to accelerate convergence, MaxPooling2D layers, a Dense layer of 128 units, and a Dropout layer (rate=0.5) to prevent overfitting.

4. Model Performance & Recommendation

Model Architecture	Train Accuracy	Val Accuracy	Test Accuracy
Model 1: Simple MLP	0.9689	0.9676	0.9709
Model 2: Simple CNN	0.9731	0.9755	0.9742
Model 3: Regularized CNN	0.9782	0.9869	0.9870



Model Recommendation: We recommend **Model 3: Regularized CNN** as the final model. It achieved the highest test accuracy of **0.9870**. Even though Model 1 and Model 2 trained faster, Model 3 is significantly more robust. The inclusion of Batch Normalization stabilized learning, and the Dropout layer (50%) acted as a strong regularizer, closing the generalization gap between training and validation accuracy to prevent overfitting.

5. Key Findings, Limitations & Next Steps

Key Findings:

1. **Spatial Convolutions are Critical:** CNNs (Models 2 and 3) outperformed the MLP (Model 1) because the 3x3 kernels preserve spatial correlations within the 2D image matrices, whereas flattening images into 1D vectors loses structural context.
2. **Dropout Directly Reduces Overfitting:** During training, Model 1 and Model 2 showed signs of validation loss flattening while training loss continued to fall. Model 3, by utilizing Dropout, forced the network to learn redundant representations, resulting in superior generalization.

Model Limitations & Future Work:

- **Sensitivity to Translation:** The current model is trained on centered digits. If digits are shifted or scaled, accuracy may degrade.
- **Proposed Improvements (Next Steps):**
 1. **Data Augmentation:** Implement random rotations, shifts, and zooms during training to build translation invariance.
 2. **Advanced Architectures:** Transition to a pre-trained network (like ResNet) or utilize hyperparameter tuning (learning rate scheduling, Adam optimizer adjustments) to push accuracy past 99.5%.
 3. **Unsupervised Pre-training:** Explore autoencoders to learn latent feature representations before classification.